

Harsh Singh

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Summary

Motivated Data Science and AI enthusiast with hands-on experience building end-to-end machine learning pipelines, retrieval-augmented generation (RAG) systems, and data-driven full-stack applications. Proficient in Python, statistical modeling, NLP, and vector search. Passionate about transforming raw data into actionable insights and deploying intelligent systems that solve real-world problems. Seeking a Data Scientist role to apply ML/AI expertise in a high-impact engineering environment.

Technical Skills

Programming Languages: Python, C, C++, JavaScript, SQL, HTML, CSS

Machine Learning & AI: Scikit-learn, FAISS, Sentence Transformers, Retrieval-Augmented Generation (RAG), NLP, Large Language Models (LLMs), Semantic Search, Vector Embeddings, Prompt Engineering

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Chart.js, Statistical Modeling, Exploratory Data Analysis (EDA)

Frameworks & Libraries: React.js, Flask, Node.js, Express.js, Tailwind CSS

Databases & Tools: MySQL, MongoDB, Git, GitHub, VS Code

Core Concepts: Data Structures & Algorithms, DBMS, Operating Systems, OOP, Computer Networks, REST APIs, JWT Authentication, Software Testing, Manual Testing, Database Testing, Usability Testing

Projects

AskPDF – AI-Powered PDF Chat Application | Flask, React.js, FAISS, Sentence Transformers

- Architected an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline integrating **FAISS** vector store and **Sentence Transformers** to enable efficient semantic search and question-answering over large, multi-document PDF corpora.
- Engineered advanced **document chunking strategies** and optimized **vector embedding** generation, significantly improving retrieval precision and reducing system latency for large-scale document sets.
- Developed a high-performance full-stack application with a **React.js** ChatGPT-style UI featuring real-time **streaming responses**, demonstrating applied NLP and LLM integration skills.
- Implemented **granular source tracking** and citation functionality, enabling verifiable AI-generated responses with direct navigation to relevant PDF pages — critical for trustworthy data-driven outputs.

WealthWave – Smart Personal Finance Manager | React, Node.js, Express, MongoDB, Chart.js

- Designed and developed a full-stack **financial analytics platform** enabling real-time tracking of income, expenses, budgets, and recurring transactions through an interactive data dashboard.
- Built comprehensive **data visualization and reporting features** including category-wise charts, monthly trend analysis, and **CSV/PDF export** capabilities, demonstrating applied data analysis and BI skill.
- Integrated an **AI-powered finance assistant** capable of answering domain-specific finance queries, leveraging optional **OpenAI API** support alongside a local insight inference mode.
- Implemented **real-time budget monitoring with automated notifications**, enabling data-driven financial decision-making and demonstrating end-to-end ML/AI feature deployment in production.
- Engineered secure **JWT-based authentication** with bcrypt password hashing, protected API routes, and custom validation rules, ensuring enterprise-grade data security.

Education

GH Raisoni College of Engineering and Management <i>B.Tech in Information Technology CGPA: 8.21</i>	Pune, India <i>Aug 2023 – Present</i>
KV No-2 Dhar Road,Udhampur <i>Higher Secondary Certificate (HSC) — Science Stream 76%</i>	India <i>2021 – 2023</i>
KV No-2 Dhar Road,Udhampur <i>Secondary School Certificate (SSC) 81%</i>	India <i>2020 – 2021</i>

Certifications & Learning

- **Self-directed ML/AI learning:** RAG systems, LLM integration, FAISS vector search, Sentence Transformers, Prompt Engineering, OpenAI API
- **Data Science Core:** Statistical modeling, EDA, Pandas/NumPy, data pipeline design

Achievements & Highlights

- Built and deployed a **production-grade RAG pipeline** (AskPDF) integrating FAISS semantic search with real-time LLM streaming — a stack aligned with industry-standard AI/ML engineering practices.
- Designed a **multi-modal AI analytics assistant** (WealthWave) combining structured financial data analysis with LLM-powered natural language querying.
- Consistently maintains a **CGPA of 8.21** while independently building complex ML and full-stack AI systems.

Research Interests

- Large Language Models (LLMs), Retrieval-Augmented Generation, Applied NLP, Generative AI, Data-centric AI Systems, Semantic Search, Statistical Machine Learning

